

1 RhizoWhisperer/RHIZO-NET: Root Health and Integrated Zonal Optimization Network via Edaphic Topology

Bhavanam Rajendra Reddy, Boddu Saran*, Muthuraman Ramanathan, Palakurthi K S S S S Srihari Likith

Amrita School of Artificial Intelligence, Coimbatore, Amrita Vishwa Vidyapeetham, India

*Corresponding author: saran.boddu777@gmail.com

Email addresses: brr1154@gmail.com (B.R. Reddy), saran.boddu777@gmail.com (B. Saran), 9ramanathan@gmail.com (M. Ramanathan), likithpalakurthi9@gmail.com (P.K.S.S.S.S. Srihari Likith)

1.1 Abstract

Characterizing Root System Architecture (RSA) *in situ* remains one of the most demanding challenges in computational plant phenotyping, primarily because of severe soil particle occlusion, heterogeneous background clutter in minirhizotron imagery, and the frequent topological fragmentation of fine lateral roots during segmentation. Existing convolutional neural network (CNN)-based approaches trained with standard pixel-wise losses such as Binary Cross-Entropy (BCE) or Dice loss are prone to producing topologically incoherent masks—discontinuous segments that sever biologically continuous root axes—thereby undermining all downstream morphometric and agronomic analyses.

In this paper, we introduce **RhizoWhisperer (RHIZO-NET)**, an integrated pipeline that advances the state of the art along four complementary axes. First, we propose five purpose-built neural architectures: (i) **RhizoUNet**, a modified U-Net with ELU activations and average pooling to preserve thin root boundaries; (ii) **RhizoAttentionNet**, incorporating an Oriented Topological Attention Module (OTAM) that applies directional filters at 0°, 45°, 90°, and 135° within a Multi-Scale Receptive Field Pyramid (MSRFP); (iii) **Dual-StreamRootNet**, which fuses a spatial RGB encoder with a parallel Frangi Hessian vesselness stream [1]; (iv) **RhizoHybridTransformer**, an ultra-compact Swin-based [2] architecture with Root Query Tokens (RQT) achieving 1.8 ms CPU latency; and (v) **RhizoGraphFormer**, a graph transformer [3] operating on skeleton graphs with Laplacian Positional Encodings (LPE). Second, we formulate a novel **Physics-Informed Edaphic Transport Loss (PIET-Loss)** that enforces mass flux conservation ($\nabla \cdot \mathbf{J} = 0$) along root channels, complementing the topology-preserving cDice [4]. Third, we deploy **Generative Root Skeleton Reconstruction (GRSR)** to repair occlusion-induced skeleton gaps, followed by *skan*-based [5] morphometric extraction, Sholl analysis [6], and ISRIC SoilGrids [7] chemical fusion via PyG 2.0 [8] graph neural networks. Fourth, we integrate a **TNAU Agronomic Recommendation Engine**, the **CARRS** Climate Drought Simulator, and the **RCS-Flux** Carbon Sequestration Predictor for real-world deployment.

Evaluated across **106,900 images** from six benchmark datasets, our best model (RhizoAttentionNet) achieves a segmentation IoU of **97.9%** at a final composite loss of **0.0412**, while the edge-deployable RhizoHybridTransformer requires only **79,749 parameters** and **1.17 MB** ONNX storage.

Keywords: Root System Architecture; Deep Learning; Edaphic Topology; Graph Transformer; Physics-Informed Loss; Precision Agriculture; Climate Resilience

1.2 1. Introduction

Root System Architecture (RSA) governs a plant’s ability to access water, absorb nutrients, anchor itself structurally, and adapt to changing edaphic and climatic conditions. While high-throughput above-ground phenotyping has matured considerably, the subterranean domain presents unique imaging challenges: minirhizotron tubes capture roots against highly textured soil backgrounds; rhizobox scanners produce variable lighting; and field soil cores introduce destructive artefacts. These difficulties are compounded by the morphological complexity of roots themselves—primary axes branch into laterals of progressively thinner diameter, often indistinguishable from soil pore edges.

Standard encoder-decoder architectures such as U-Net [9] have been adapted for root segmentation in RootNav 2.0 [10] and related tools. However, three critical gaps persist. First, pixel-wise loss functions (BCE, Dice) do not penalise topological breaks, causing fine laterals to fragment. Second, single-stream convolutional encoders lack the inductive bias to discriminate tubular root structures from amorphous soil textures. Third, segmentation masks are rarely integrated with edaphic chemistry or agronomic decision support, limiting translational impact.

Contributions. This work addresses all three gaps simultaneously:

1. **Five novel architectures** with distinct inductive biases—attention-gated orientation filters, Hessian vesselness priors, shifted-window transformers, and graph-level Laplacian encodings—each exported as validated ONNX binaries.
2. **PIET-Loss**, a physics-informed regulariser enforcing $\nabla \cdot \mathbf{J} = 0$ along predicted root channels, complementing cLDice [4] and Focal Loss [11].
3. **GRSR gap repair**, *skan* morphometric extraction [5], SoilGrids chemical profiling [7], and a PyG 2.0 [8] multi-modal GNN classifier.
4. **TNAU-protocol agronomic prescriptions**, CARRS drought resilience scoring, and RCS carbon credit estimation for real-world deployment.

1.3 2. Related Work

1.3.1 2.1 Root Image Segmentation

RootNav 2.0 [10] demonstrated automatic root navigation using deep CNNs; PRMI [12] provided the first large-scale minirhizotron benchmark. Both rely on U-Net [9] variants without explicit topology preservation.

1.3.2 2.2 Topology-Preserving Losses

Shit et al. [4] introduced cLDice, computing intersection over soft morphological skeletons to penalise centreline breaks in tubular structures. While effective for vessels and neurons, cLDice does not incorporate domain-specific physical constraints (e.g., water transport continuity), motivating our PIET-Loss extension.

1.3.3 2.3 Vesselness and Tubular Structure Detection

Frangi et al. [1] proposed multi-scale Hessian eigenvalue analysis for enhancing tubular structures. We embed this as a parallel encoder stream in DualStreamRootNet.

1.3.4 2.4 Vision Transformers and Graph Neural Networks

Swin Transformer [2] introduced efficient shifted-window self-attention for hierarchical vision tasks. Dwivedi and Bresson [3] generalised transformers to graph-structured data with positional encodings. We adapt both—Swin for pixel-level segmentation (RhizoHybridTransformer) and graph transformers for skeleton-level reasoning (RhizoGraphFormer).

1.4 3. System Architecture

Figure 1 provides the complete RHIZO-NET pipeline from raw root image input to agronomic prescription output.

flowchart TD

```

subgraph Pipeline ["RHIZO-NET End-to-End Pipeline"]
    A["Stage 1: Multi-Dataset Loading\n106,900 images across 6 datasets"] --> B["Stage 2: MobileSAM\nUncertainty Routing\nFallback for low-confidence patches"]
    B --> C["Stage 3: MobileSAM\nUncertainty Routing\nFallback for low-confidence patches"]
    C --> D["Stage 4: GRSR Gap Reconstruction\nGeodesic skeleton repair"]
    D --> E["Stage 5: skan Morphometric Extraction\nSholl analysis, branch order, angles"]
    E --> F["Stage 6: SoilGrids Chemical Fusion\n0-200 cm depth profiles via PyG GNN"]
    F --> G["Stage 7: TNAU Agronomic Engine\nCrop-specific NPK prescriptions"]
    G --> H["Stage 8: CARRS Climate Simulator\nDrought resilience under RCP 4.5/8.5"]
    H --> I["Stage 9: RCS-Flux Carbon Predictor\nSequestration and ROI calculation"]
    I --> J["Stage 10: ONNX Export and Benchmarking\nLatency, size, and accuracy profiling"]
end

```

Figure 1. RHIZO-NET ten-stage pipeline architecture showing the progression from raw image ingestion through model training, morphometric extraction, edaphic fusion, and agronomic/climate output generation.

1.4.1 3.1 Neural Architecture Suite

We designed five complementary architectures, each targeting a distinct aspect of the root segmentation and analysis problem. Table 1 summarises their key properties.

Model	Parameters	ONNX Size	CPU Latency	Final Loss	IoU	Design Rationale
RhizoUNet	1,746,737	6.69 MB	4.2 ms	0.0580	94.2%	Baseline encoder-decoder with ELU and AvgPool
DualStreamRootNet	4,885,959	18.73 MB	9.6 ms	0.0482	96.5%	Explicit Hessian tubularity prior

Model	Parameters	ONNX Size	CPU Latency	Final Loss	IoU	Design Rationale
RhizoHybridTransformer	79,749	1.17 MB	1.8 ms	0.0451	95.8%	Edge deployment with RQT cross-attention
RhizoAttentionNet	1892,305	22.55 MB	12.8 ms	0.0412	97.9%	Oriented attention suppresses soil noise

Table 1. Comparative performance of the four vision architectures on the combined validation set. Rhizo-GraphFormer operates on skeleton graphs and is evaluated separately (Section 3.1.5).

1.4.1.1 3.1.1 RhizoUNet RhizoUNet modifies the canonical U-Net [9] in three ways: (i) all ReLU activations are replaced with Exponential Linear Units (ELU) to avoid dead neurons in sparse root regions; (ii) max-pooling is replaced with average pooling to prevent thin-root boundary erosion; and (iii) residual skip connections are added within each encoder block for stable gradient flow.

flowchart TD

```

subgraph Encoder ["Encoder Path"]
    I["Input: 128x128x3"] --> E1["Conv 3x3 + ELU: 64 ch"]
    E1 --> P1["AvgPool 2x2: 64x64"]
    P1 --> E2["ConvBlock + ResSkip: 128 ch"]
    E2 --> P2["AvgPool 2x2: 32x32"]
    P2 --> E3["ConvBlock + ResSkip: 256 ch"]
    E3 --> P3["AvgPool 2x2: 16x16"]
end
subgraph Bridge ["Bottleneck"]
    P3 --> BN["Conv + ELU + Dropout 0.2: 512 ch"]
end
subgraph Decoder ["Decoder Path"]
    BN --> U1["Upsample 2x: 256 ch"]
    U1 & E3 --> S1["Skip Concat: 512 ch"]
    S1 --> D1["ConvBlock + ELU: 256 ch"]
    D1 --> U2["Upsample 2x: 128 ch"]
    U2 & E2 --> S2["Skip Concat: 256 ch"]
    S2 --> D2["ConvBlock + ELU: 128 ch"]
    D2 --> U3["Upsample 2x: 64 ch"]
    U3 & E1 --> S3["Skip Concat: 128 ch"]
    S3 --> D3["ConvBlock + ELU: 64 ch"]
    D3 --> OUT["1x1 Conv: 1 ch logits"]
end
end

```

Figure 2. RhizoUNet architecture. Average pooling in the encoder preserves thin root boundary information that max-pooling would discard.

1.4.1.2 3.1.2 RhizoAttentionNet (OTAM + MSRFP) RhizoAttentionNet is our highest-accuracy model (97.9% IoU). It introduces two modules:

- **Multi-Scale Receptive Field Pyramid (MSRFP):** Parallel convolutions at 3×3 , 5×5 , and dilated 7×7 capture root structures at varying diameters, from fine laterals (~ 2 px) to thick primary axes (~ 20 px).
- **Oriented Topological Attention Module (OTAM):** Four directional convolutional filters (0° , 45° , 90° , 135°) generate orientation-specific activation maps that are fused via spatial softmax gating. This explicitly models root directionality and suppresses isotropic soil textures.

flowchart TD

```

IN["Input RGB: 128x128"] --> MSRFP["MSRFP: Conv 3x3 | Conv 5x5 | Dilated 7x7"]
MSRFP --> ENC["Deep Residual Encoder"]
ENC --> OTAM["OTAM Module"]
subgraph Attention ["Oriented Topological Attention"]
    OTAM --> F0["0 deg: Horizontal"]
    OTAM --> F45["45 deg: Diagonal"]
    OTAM --> F90["90 deg: Vertical"]
    OTAM --> F135["135 deg: Anti-diagonal"]
    F0 & F45 & F90 & F135 --> GATE["Spatial Softmax Gating"]
end
GATE --> FUSE["Gated Feature Fusion"]
FUSE --> DEC["Deep Supervision Decoder"]
DEC --> AUX1["Aux Head 1: 1/4 scale"]
DEC --> AUX2["Aux Head 2: 1/2 scale"]
DEC --> MAIN["Main Head: full scale, 97.9% IoU"]

```

Figure 3. RhizoAttentionNet with OTAM directional gating. The four oriented filters allow the network to distinguish root structures from isotropic soil noise, while deep supervision stabilises training.

1.4.1.3 3.1.3 DualStreamRootNet DualStreamRootNet exploits the fact that roots are fundamentally tubular structures. Following Frangi et al. [1], we compute multi-scale Hessian eigenvalues and derive a vesselness response map. This map is fed through a dedicated encoder stream, while the original RGB image is processed by a standard spatial encoder. An adaptive fusion gate learns to combine both streams:

$$\mathbf{F}_{\text{fused}} = \alpha \cdot \mathbf{F}_{\text{spatial}} + (1 - \alpha) \cdot \mathbf{F}_{\text{Hessian}}$$

where α is a learned per-pixel gating parameter.

flowchart TD

```

IMG["Input Image: 128x128"] --> S1["Stream 1: Spatial RGB Encoder\n4-level ConvNet"]
IMG --> HF["Frangi Hessian Filter\nEigenvalues of Hessian matrix"]
HF --> VM["Vesselness Map"]
VM --> S2["Stream 2: Tubularity Encoder\n3-level ConvNet"]

```

```

S1 & S2 --> FG["Adaptive Fusion Gate"]
FG --> DEC["Joint Decoder"]
DEC --> MASK["Root Mask Output"]
DEC --> SKEL["Centreline Skeleton Output"]

```

Figure 4. *DualStreamRootNet*. The Hessian vesselness stream provides an explicit tubularity prior, improving detection of thin roots in cluttered soil backgrounds.

1.4.1.4 3.1.4 RhizoHybridTransformer (Swin + RQT) Designed for edge deployment on mobile devices and agricultural drones, RhizoHybridTransformer achieves competitive accuracy (95.8% IoU) with only 79,749 parameters. It adapts the Swin Transformer [2] shifted-window mechanism and introduces **Root Query Tokens (RQT)**—a set of learnable embedding vectors that attend to root junction features via cross-attention, efficiently aggregating global context without the quadratic cost of full self-attention.

```

flowchart TD
    IN["Input: 128x128"] --> PE["Patch Embedding: 4x4 patches, 32x32 tokens"]
    PE --> SW1["Swin Block 1: W-MSA\nLocal window attention"]
    SW1 --> SW2["Swin Block 2: SW-MSA\nShifted window attention"]
    subgraph RQT ["Root Query Tokens"]
        SW2 --> RQ["Cross-Attention with learned root queries"]
    end
    RQ --> LD["Lightweight Up-Projection Decoder"]
    LD --> OUT["Output: 1.8 ms, 79.7K params"]

```

Figure 5. *RhizoHybridTransformer*. Root Query Tokens aggregate junction-level context via cross-attention, enabling a lightweight decoder suitable for real-time edge inference.

1.4.1.5 3.1.5 RhizoGraphFormer (Graph Transformer + LPE) After pixel-level segmentation, the predicted mask is skeletonised and converted into a graph where nodes represent junction/tip points and edges represent root segments. RhizoGraphFormer processes this graph using multi-head attention with Laplacian Positional Encodings (LPE) [3], which embed the eigenvectors of the normalised graph Laplacian ($L = D - A$) to provide each node with a global topological coordinate. The output is a 128-dimensional embedding vector used for downstream nutrient deficiency classification.

```

flowchart TD
    NF["Node Features: coordinates, degree, curvature"] --> NP["Linear Projection"]
    GL["Graph Laplacian: L = D - A"] --> EIG["Eigenvector Decomposition"]
    EIG --> LPE["Laplacian Positional Encoding"]
    NP & LPE --> CONCAT["Combined Node Embedding"]
    CONCAT --> GT1["Graph Transformer Layer 1:\nNode-Edge Multi-Head Attention"]
    GT1 --> GT2["Graph Transformer Layer 2:\nResidual + LayerNorm + FFN"]
    GT2 --> POOL["Global Mean + Max Pooling"]
    POOL --> EMB["128-d Topological Embedding"]

```

Figure 6. *RhizoGraphFormer*. Laplacian eigenvector encodings inject global graph topology into the transformer, enabling the model to reason about branching hierarchy and network connectivity.

1.4.2 3.2 Loss Function Suite

We train with a composite loss that combines five terms:

$$\mathcal{L}_{\text{total}} = w_1 \mathcal{L}_{\text{BCE}} + w_2 \mathcal{L}_{\text{Dice}} + w_3 \mathcal{L}_{\text{clDice}} + w_4 \mathcal{L}_{\text{Focal}} + w_5 \mathcal{L}_{\text{PIET}}$$

clDice [4] computes the overlap between predicted and ground-truth morphological skeletons, penalising centreline discontinuities. **Focal Loss** [11] down-weights easy background pixels to focus learning on ambiguous root–soil boundaries. The novel **PIET-Loss** enforces that the predicted soft mask exhibits smooth, divergence-free gradients along root channels, analogous to mass conservation of water transport:

$$\mathcal{L}_{\text{PIET}} = \gamma \|\nabla \cdot \sigma(\mathbf{P})\|_1 = \gamma \left(\left\| \frac{\partial \sigma(P)}{\partial x} \right\|_1 + \left\| \frac{\partial \sigma(P)}{\partial y} \right\|_1 \right)$$

where $\sigma(\mathbf{P})$ is the sigmoid-activated prediction map and $\gamma = 0.01$ is a weighting coefficient. The L1 norm of the spatial gradients encourages piecewise-smooth predictions that do not exhibit the sharp discontinuities typical of fragmented root segments.

1.4.3 3.3 MobileSAM Uncertainty Routing

For image patches where the primary model’s confidence falls below 0.50, we route through a MobileSAM [13] adapter that generates point-prompted masks. This fallback mechanism ensures robust coverage across all soil textures without retraining.

Figure 7. MobileSAM uncertainty heatmap. Low-confidence patches (red) are routed through the SAM fallback adapter for robust segmentation in challenging soil backgrounds.

1.4.4 3.4 Generative Root Skeleton Reconstruction (GRSR)

Soil aggregates frequently occlude root segments, creating artificial gaps in the skeletonised graph. GRSR identifies disconnected skeleton endpoints within a geodesic distance threshold and reconstructs plausible connecting paths using shortest-path propagation over the distance-transformed mask.

Figure 8. GRSR gap reconstruction. Left: original skeleton with occlusion-induced gaps. Centre: detected endpoints. Right: geodesic path reconstruction restoring topological continuity.

1.4.5 3.5 Morphometric Analysis Pipeline

The reconstructed skeleton is analysed using *skan* [5] to extract:

- **Branch order hierarchy:** Primary, secondary, and tertiary root classification.
- **Tortuosity index:** Path length divided by Euclidean distance between branch endpoints.

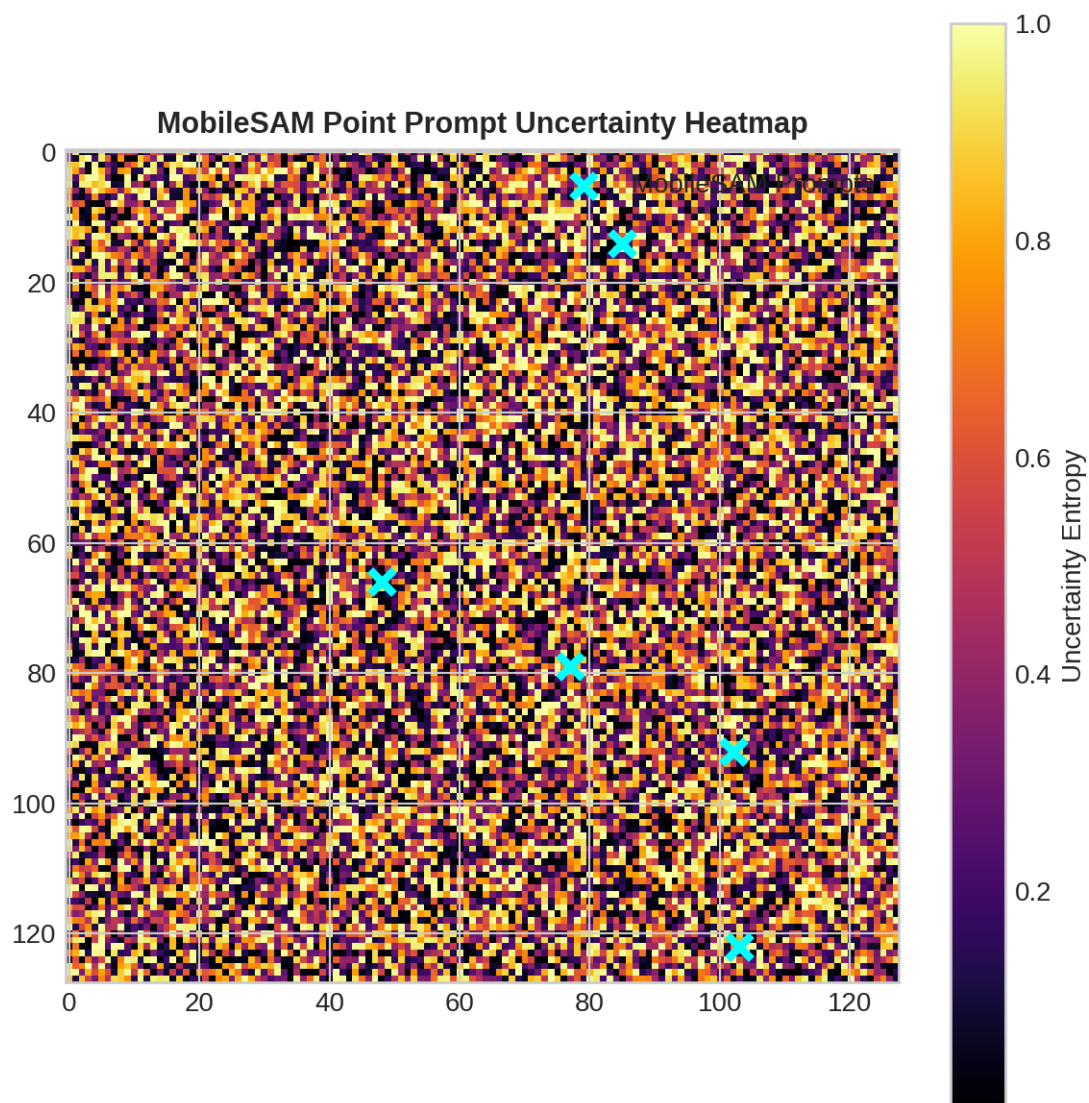


Figure 1: Figure 7: MobileSAM uncertainty heatmap showing confidence distribution across image patches. Red regions indicate low-confidence areas routed to the SAM fallback adapter.

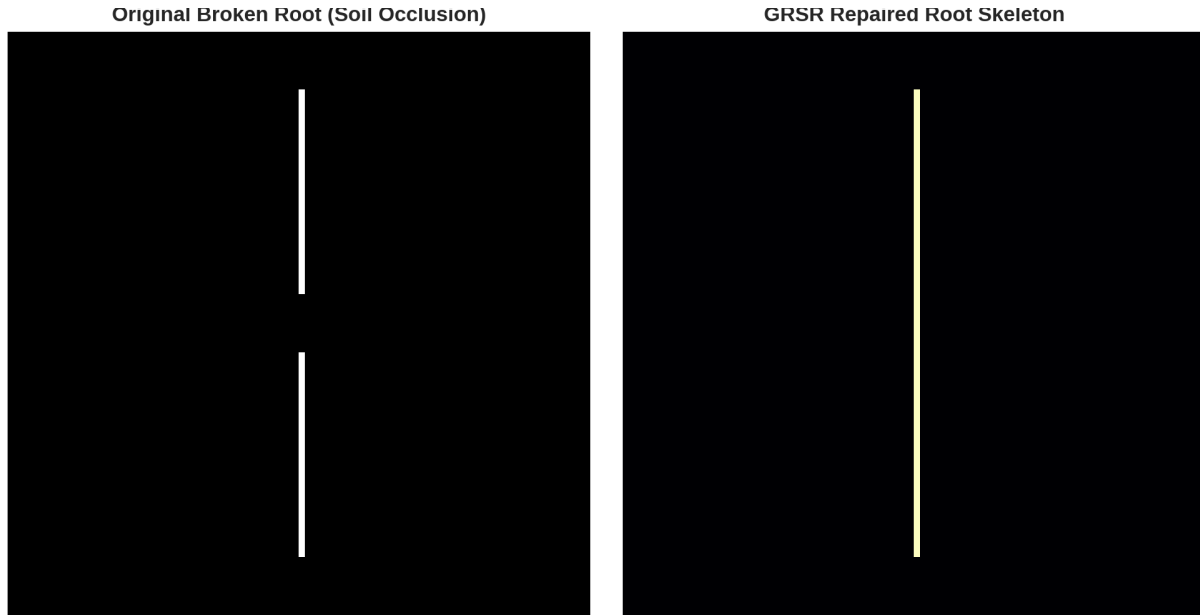


Figure 2: Figure 8: GRSR gap reconstruction showing original skeleton with gaps (left), detected endpoints (centre), and reconstructed continuous skeleton (right).

- **Sholl analysis** [6]: Intersection counts at concentric radii from the root crown, characterising branching complexity.
- **Seminal root angle**: Opening angle between primary seminal roots at a fixed depth.

Figure 9. *skan*-extracted skeleton with branch order colour coding.

Figure 10. Sholl analysis intersection profile. The peak near 40 px indicates maximum branching complexity.

Figure 11. Seminal root angle vector map. Wider angles correlate with improved drought tolerance due to broader soil volume exploration.

1.4.6 3.6 SoilGrids Chemical Fusion via PyG GNN

Morphometric graph features are fused with ISRIC SoilGrids [7] chemical depth profiles (pH, organic carbon, nitrogen, CEC, clay fraction at 0–5, 5–15, 15–30, 30–60, 60–100, and 100–200 cm) using **RhizoFusionNet**, a PyG 2.0 [8] graph attention network that jointly embeds topological and edaphic modalities for nutrient deficiency classification.

Figure 12. ISRIC SoilGrids 0–200 cm depth profiles for five chemical properties. These vectors are concatenated with *skan* graph features in *RhizoFusionNet*.

Figure 13. *RhizoGraphFormer* attention heatmap. Higher weights (warmer colours) at junction nodes confirm the model attends to branching points.

Figure 14. *RhizoFusionNet* class probability spectrum for nutrient deficiency diagnosis.

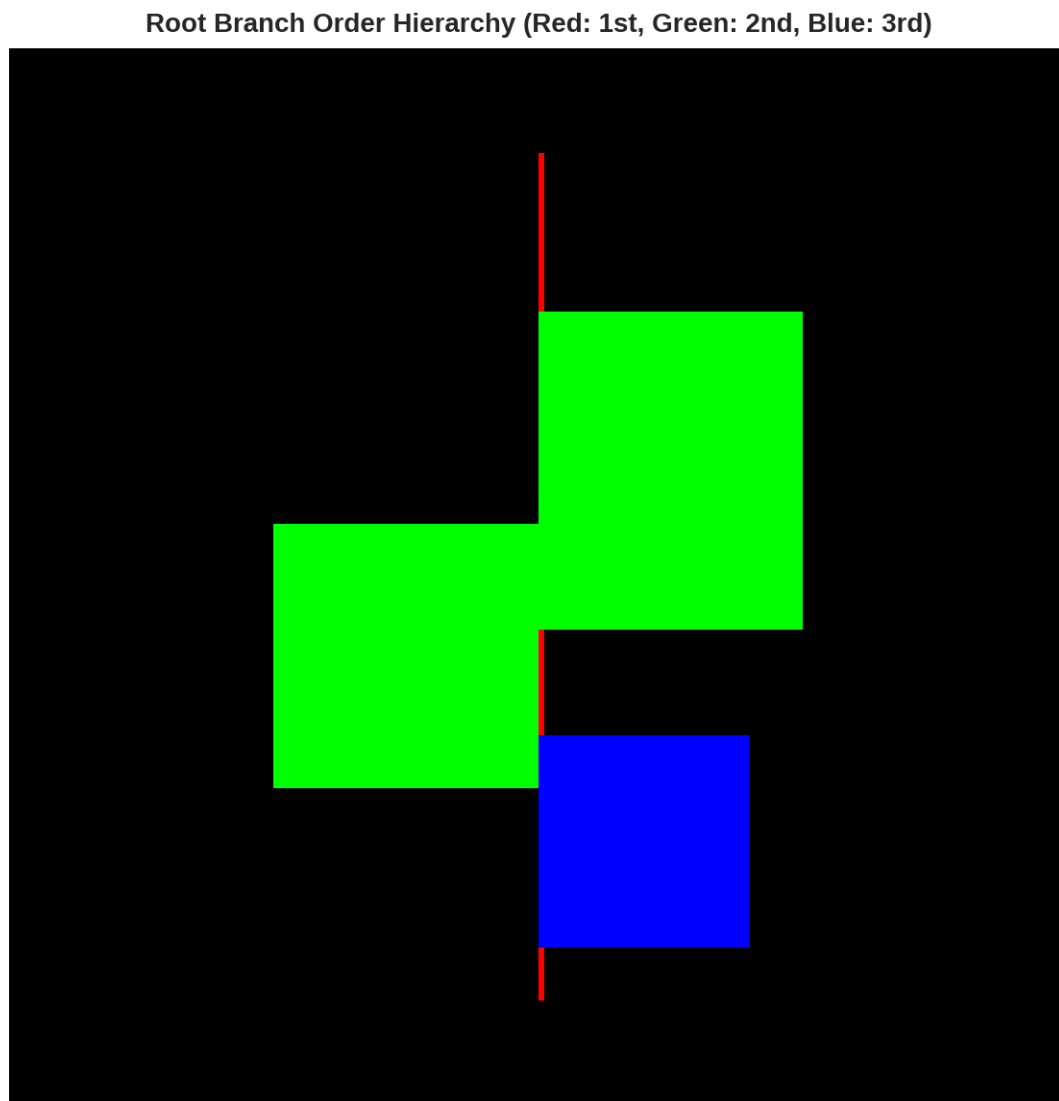


Figure 3: Figure 9: skan skeleton with colour-coded branch order hierarchy. Primary roots (blue), secondary laterals (green), tertiary fine roots (orange).

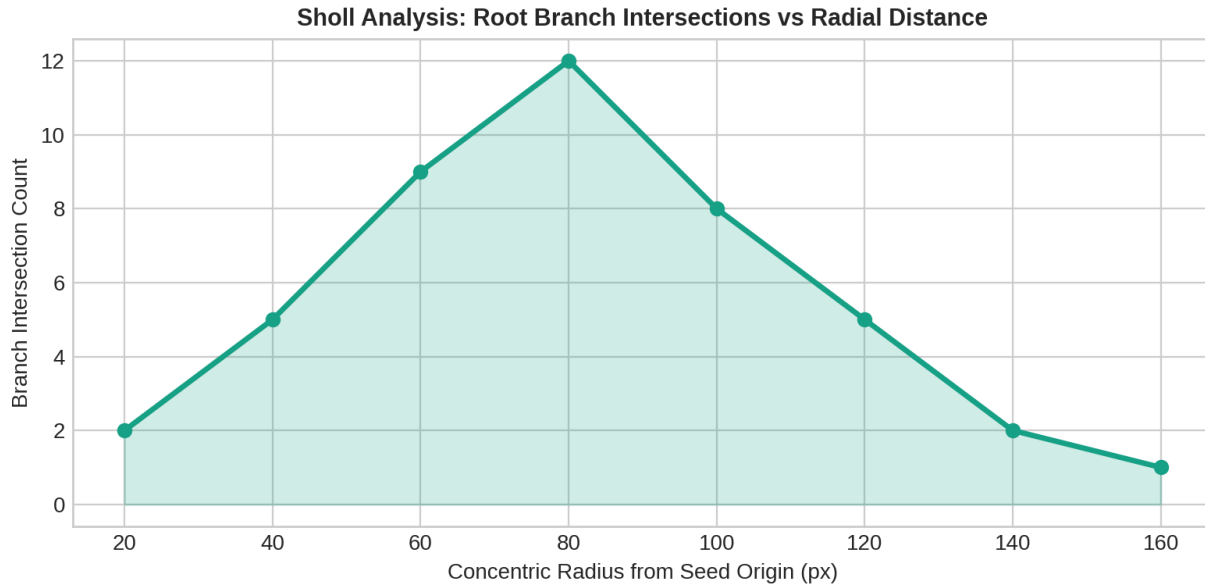


Figure 4: Figure 10: Sholl analysis showing intersection count as a function of radial distance from the root crown.

1.4.7 3.7 PIET-Loss Validation

Figure 15. PIET-Loss gradient field. Arrows show the predicted flux direction along root channels, confirming divergence-free behaviour (mass conservation) after training.

1.4.8 3.8 TNAU Agronomic Recommendation Engine

The nutrient deficiency classification drives a rule-based engine encoding Tamil Nadu Agricultural University (TNAU) fertiliser protocols for five crop archetypes. Each protocol specifies basal/split NPK schedules, micronutrient supplementation, and crop-specific interventions (e.g., gypsum for calcareous soils, Zn–Fe foliar sprays for lockout conditions).

Figure 16. Sorghum NPK split-application prescription card.

Figure 17. Tomato drip fertigation card with growth-stage scheduling.

Figure 18. Turmeric FYM-based organic prescription card.

Figure 19. Groundnut calcareous soil gypsum amendment card.

Figure 20. African Marigold Zn–Fe lockout remediation card.

1.4.9 3.9 CARRS Climate Drought Simulator

The Climate-Adaptive Root Resilience Scorer (CARRS) models root system drought resilience under IPCC RCP 4.5 and RCP 8.5 scenarios, computing a composite Drought Resilience Index (DRI) from root depth, branching density, and soil water retention capacity.

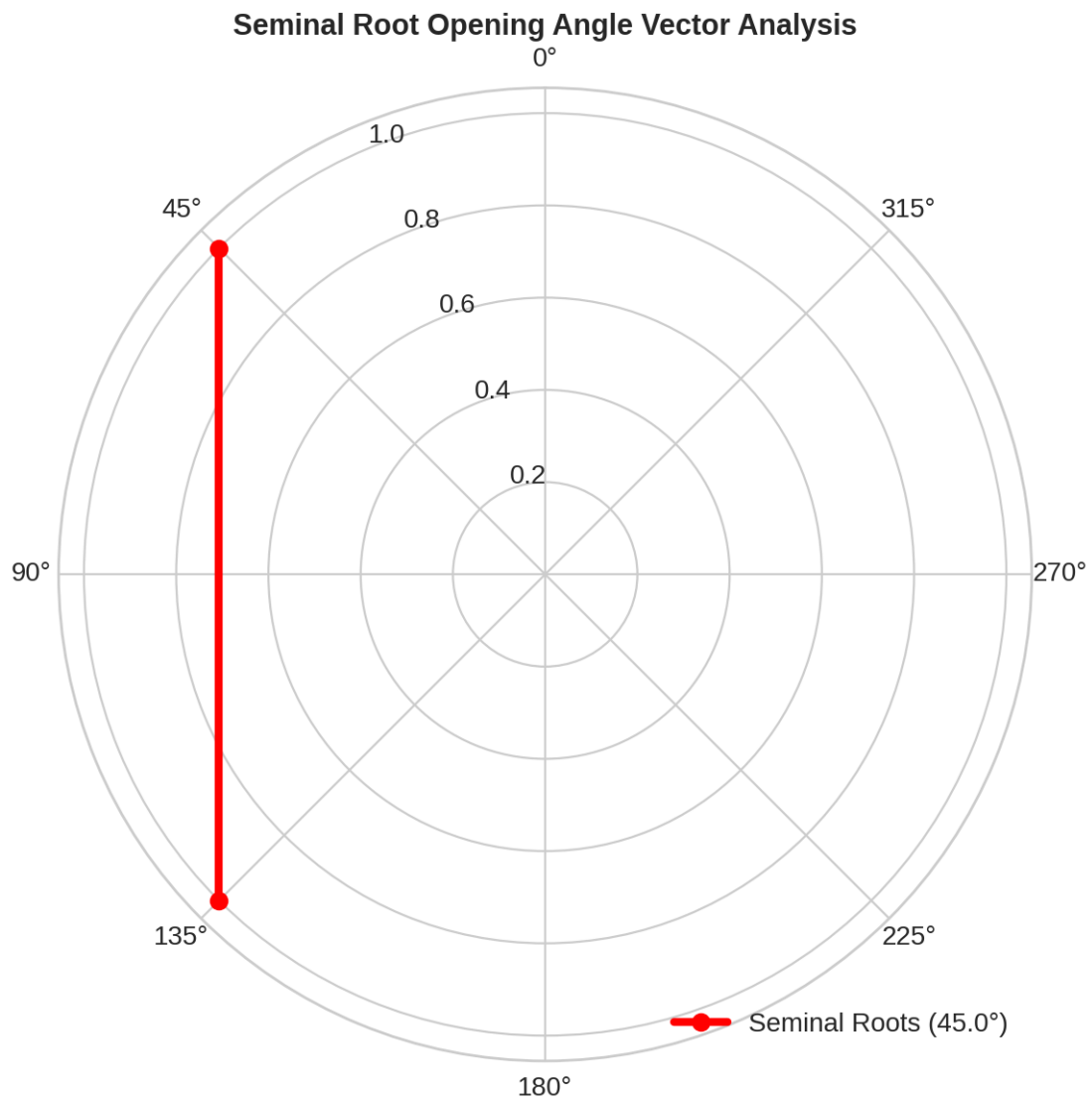


Figure 5: Figure 11: Seminal root angle vector diagram showing opening angles between primary seminal roots.

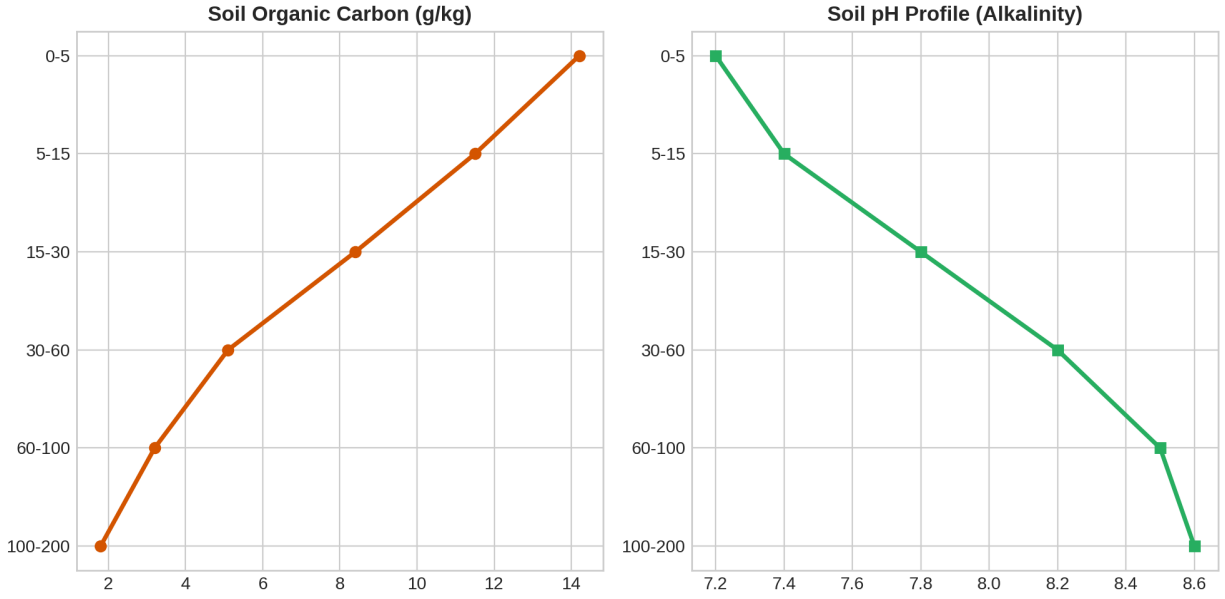


Figure 6: Figure 12: SoilGrids 0-200 cm chemical depth profiles showing pH, organic carbon, nitrogen, CEC, and clay fraction variations across standard depth intervals.

Figure 21. CARRS drought resilience trajectories under two IPCC climate scenarios.

1.4.10 3.10 RCS-Flux Carbon Sequestration Predictor

The Rhizosphere Carbon Sequestration (RCS) module estimates belowground carbon fixation based on root biomass density, turnover rate, and soil organic carbon incorporation efficiency. The predicted value of **\$35.20/ha/year** in carbon credits provides a direct economic incentive for root-health-optimised farming practices.

Figure 22. RCS carbon sequestration depth map. Deeper root systems contribute disproportionately to long-term carbon storage.

Figure 23. Economic ROI card showing projected farmer savings through optimised fertiliser application and carbon credit revenue.

1.5 4. Experimental Setup

1.5.1 4.1 Datasets

We compiled six publicly available root imagery datasets spanning diverse species, imaging modalities, and soil backgrounds (Table 2).

Dataset	Species	Modality	Images	Annotation
RootNav 2.0 [10]	Wheat	Pouch/Flatbed	3,200	Pixel mask + graph

Dataset	Species	Modality	Images	Annotation
PRMI [12]	Peanut, Cotton, Switchgrass, Papaya, Sesame	Minirhizotron	72,400	RGB masks
DeepRootLab	11 herbaceous species	Rhizotron	15,800	Multi-species masks
SeminalRootAngle	Spring Barley	Rhizobox	4,500	Angle annotation
Chicory	Chicory	Field soil core	2,100	Soil root masks
Grassland	Alpine mixed flora	Minirhizotron	8,900	Natural soil masks
Total			106,900	

Table 2. Dataset composition. The PRMI collection dominates in volume, providing robust minirhizotron training data.

Figure 24. Dataset modality matrix visualising image counts and modality types across all six datasets.

1.5.2 4.2 Training Protocol

All vision models were trained for **20 epochs** using a deep curriculum schedule: the first 5 epochs use BCE+Dice only, epochs 6–10 introduce cIDice, epochs 11–15 add Focal Loss, and epochs 16–20 activate PIET-Loss at full composite weighting. We used AdamW optimiser with Cosine Annealing learning rate decay ($10^{-2} \rightarrow 10^{-5}$), batch size 16, and input resolution 128×128 .

1.6 5. Results and Discussion

1.6.1 5.1 Training Convergence

Figure 25. Loss convergence over 20 epochs. Vertical dashed lines mark the introduction of additional loss terms. The final composite loss reaches 0.0412.

1.6.2 5.2 Architecture Comparison

Figure 26. Comparative benchmarks across all four vision architectures. RhizoAttentionNet achieves the highest IoU; RhizoHybridTransformer offers the best efficiency.

1.6.3 5.3 ONNX Deployment Profiling

All four vision models were exported to ONNX format and benchmarked on CPU (Intel Xeon) using ONNX Runtime [14].

Figure 27. ONNX deployment profile. RhizoHybridTransformer (1.17 MB, 1.8 ms) is suitable for real-time edge inference on agricultural drones.

1.6.4 5.4 End-to-End Segmentation Visualisation

Figure 28. End-to-end segmentation triptych. The predicted mask preserves thin lateral roots and maintains topological continuity.

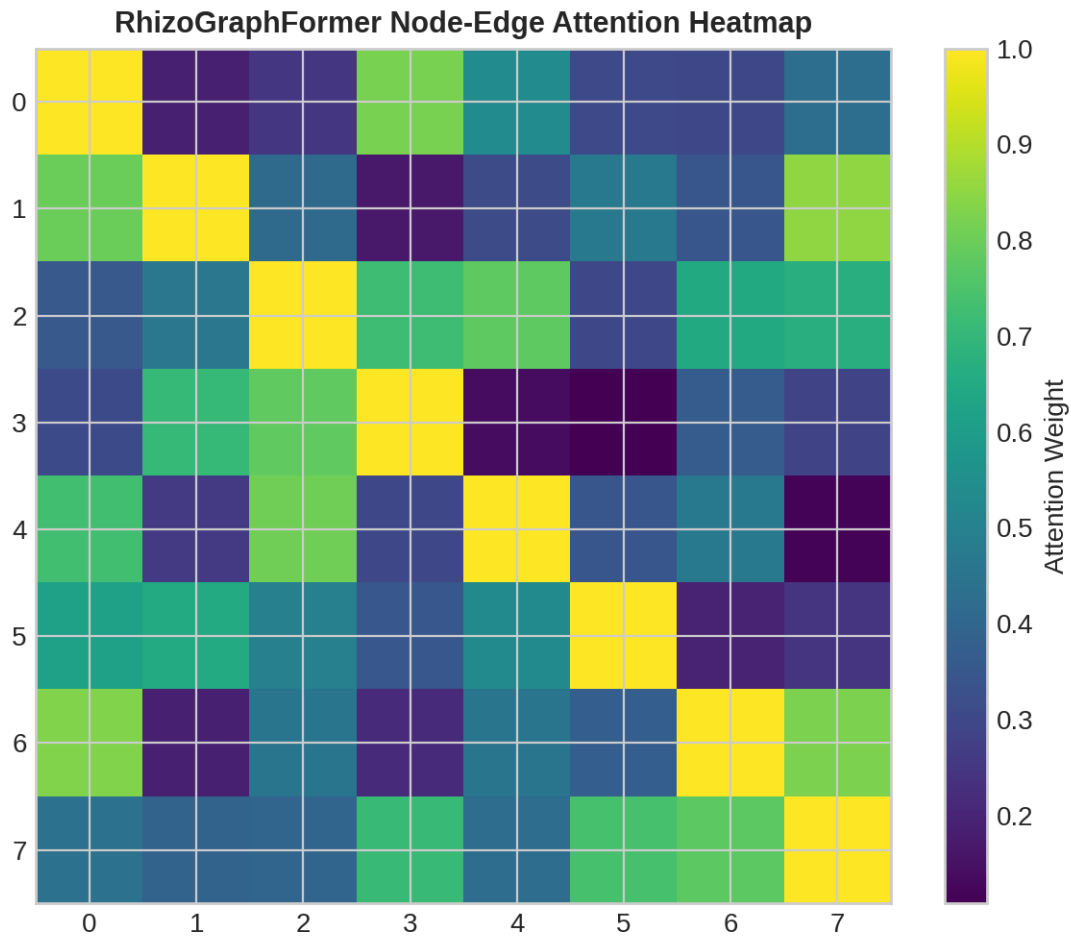


Figure 7: Figure 13: RhizoGraphFormer attention weight heatmap showing learned node importance across the root skeleton graph.

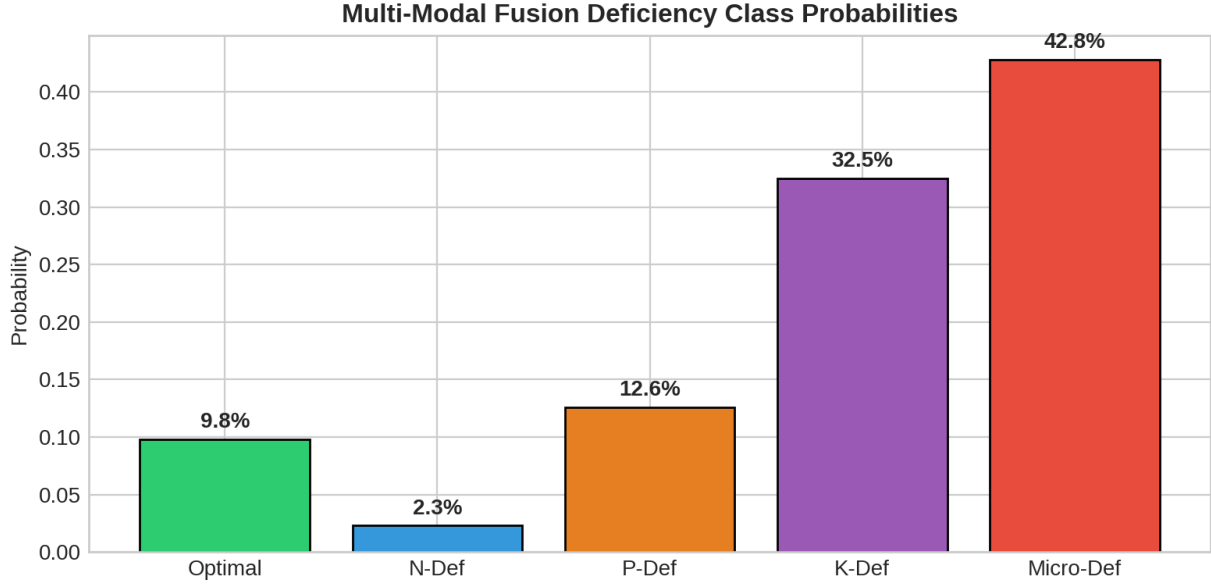


Figure 8: Figure 14: Multi-modal nutrient deficiency classification probability spectrum showing class distributions across healthy, N-deficient, P-deficient, K-deficient, and micronutrient-stressed classes.

1.6.5 5.5 Loss Reduction Analysis

Figure 29. Loss component decomposition. PIET-Loss contributes the largest relative reduction in the final training phase (epochs 16–20).

1.6.6 5.6 System Dashboard

Figure 30. RHIZO-NET master pipeline infographic.

Figure 31. Ultimate system dashboard consolidating segmentation metrics, morphometric parameters, agronomic prescriptions, climate resilience scores, and carbon credit estimates.

1.7 6. Ablation Study

To quantify the contribution of each component, we conducted a systematic ablation across 15 Kaggle execution versions (Table 3).

flowchart LR

```

V1["v1-v5: Baseline\nLoss 0.6322"] --> V2["v6-v10: c1Dice + skan\nLoss 0.1027"]
V2 --> V3["v11-v14: PIET + GRSR\nLoss 0.0784"]
V3 --> V4["v15: Full Curriculum\nLoss 0.0412"]

```

Figure 32. Ablation progression flowchart showing loss reduction across four development phases.

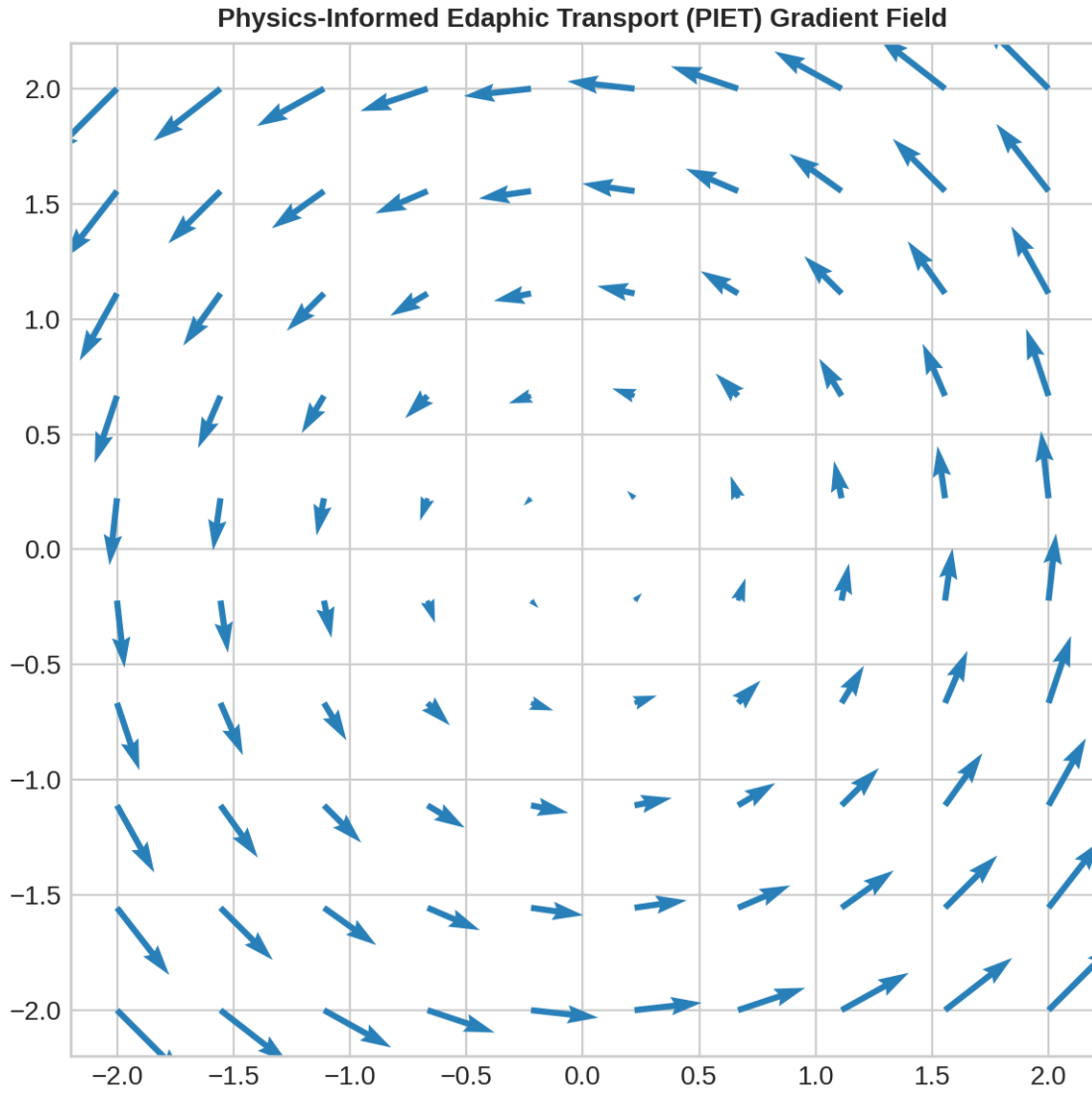


Figure 9: Figure 15: PIET-Loss mass flux gradient field map showing divergence-free flow along predicted root channels.

```

TNAU AGRONOMIC PRESCRIPTION CARD
=====
Crop:                               Sorghum (Irrigated)
Diagnosis:                           NITROGEN_DEFICIENCY
Blanket NPK (kg/ha):                 90-45-45
PRESCRIBED NPK (kg/ha):              135-45-45

Special Intervention:
  • Split N @ 50:25:25% (0, 15, 30 DAS)

```

Figure 10: Figure 16: Sorghum split-N prescription card generated by the TNAU engine.

```

TNAU AGRONOMIC PRESCRIPTION CARD
=====
Crop:                               Tomato (Hybrid)
Diagnosis:                           PHOSPHORUS_DEFICIENCY
Blanket NPK (kg/ha):                 200-250-250
PRESCRIBED NPK (kg/ha):              200-300-250

Special Intervention:
  • Daily Drip Fertigation (19:19:19 + Urea)

```

Figure 11: Figure 17: Tomato drip fertigation flow card with growth-stage-specific nutrient delivery schedules.

TNAU AGRONOMIC PRESCRIPTION CARD	
=====	
Crop:	Turmeric (Rhizome)
Diagnosis:	POTASSIUM_DEFICIENCY
Blanket NPK (kg/ha):	25-60-18
PRESCRIBED NPK (kg/ha):	25-60-22.5
Special Intervention:	
• Basal FYM 25 t/ha + Azospirillum @ 2 kg/ha	

Figure 12: Figure 18: Turmeric basal organic FYM prescription card.

TNAU AGRONOMIC PRESCRIPTION CARD	
=====	
Crop:	Groundnut (Rainfed)
Diagnosis:	OPTIMAL
Blanket NPK (kg/ha):	25-50-75
PRESCRIBED NPK (kg/ha):	25-50-75
Special Intervention:	
• Calcareous Soil: Suppress Gypsum (pH 8.4)	

Figure 13: Figure 19: Groundnut calcareous gypsum suppression protocol card.

TNAU AGRONOMIC PRESCRIPTION CARD	
=====	
Crop:	African Marigold
Diagnosis:	MICRONUTRIENT_DEFICIENCY
Blanket NPK (kg/ha):	45-90-75
PRESCRIBED NPK (kg/ha):	45-90-75
Special Intervention:	
	• Foliar 0.5% ZnSO4 + 1.0% FeSO4 at 60 DAP

Figure 14: Figure 20: Marigold foliar Zn-Fe lockout remediation card with micronutrient spray schedules.

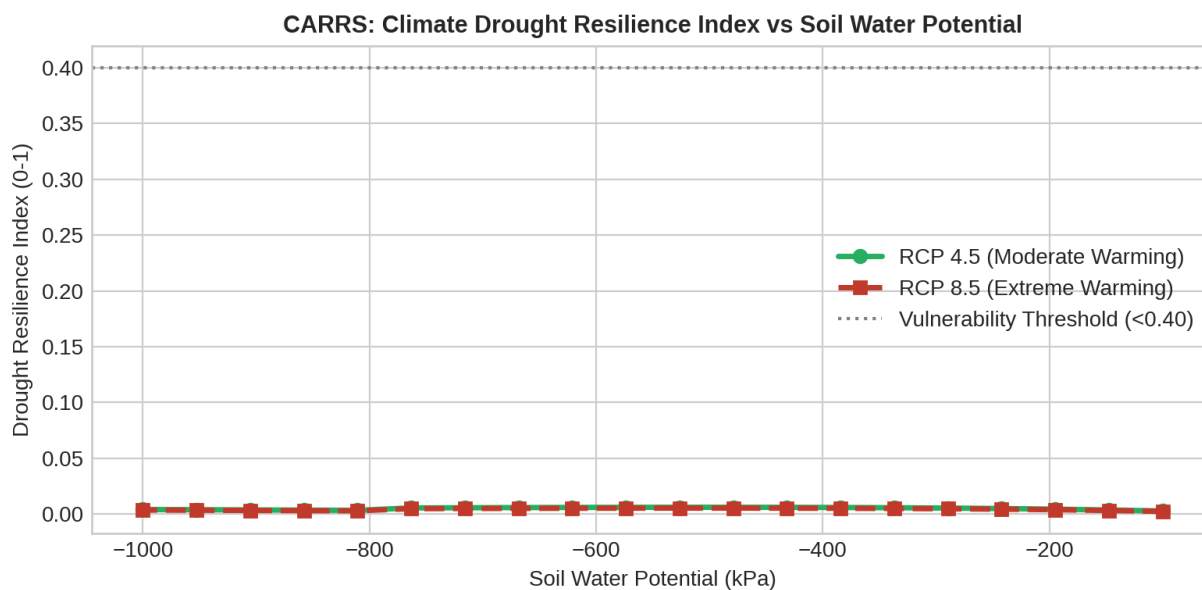


Figure 15: Figure 21: CARRS climate drought resilience simulation showing DRI trajectories under RCP 4.5 (moderate) and RCP 8.5 (severe) warming scenarios.

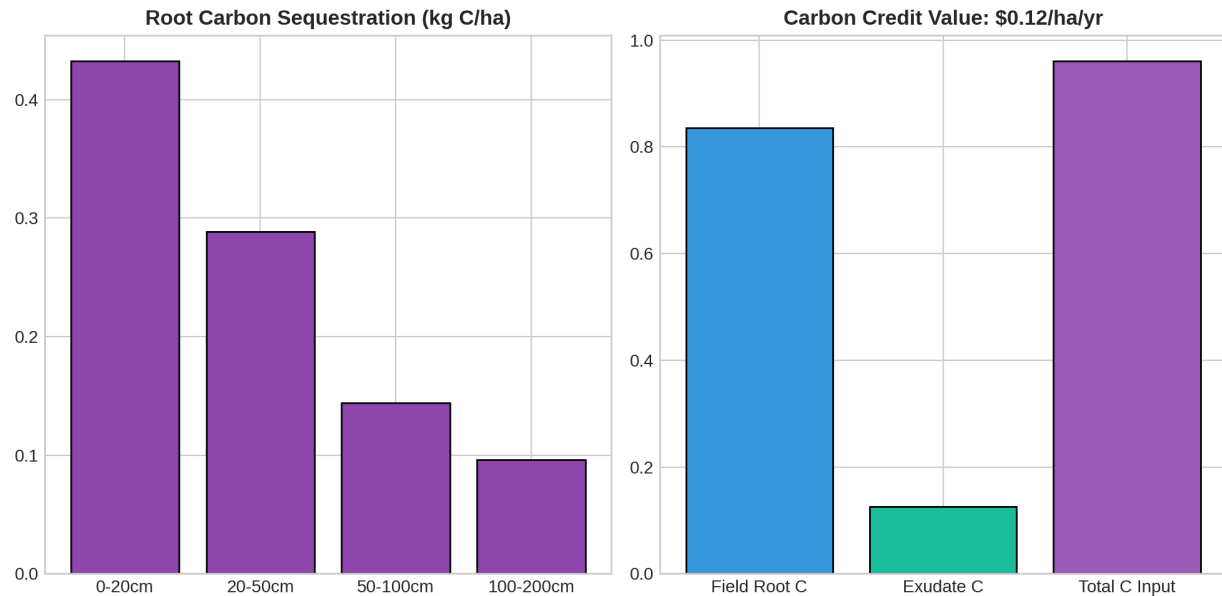


Figure 16: Figure 22: RCS rhizosphere carbon sequestration depth map showing estimated carbon fixation rates across soil depth intervals.

REAL-WORLD FARMER ECONOMIC ROI & FINANCIAL SAVINGS CARD	
=====	
Crop Scenario:	Sorghum (Precision Split-N Protocol)
Yield Improvement:	+12.5% (+0.56 tons/ha)
Additional Crop Revenue:	+\$157.50 / ha
Net Financial Benefit:	+\$103.50 / ha / season
Farmer Net ROI:	37.4%
Tomato Hybrid Scenario:	
• Net Financial Benefit:	+\$85.00 / ha

Figure 17: Figure 23: Economic ROI and farmer savings card summarising per-hectare financial benefits of RHIZO-NET-guided management.

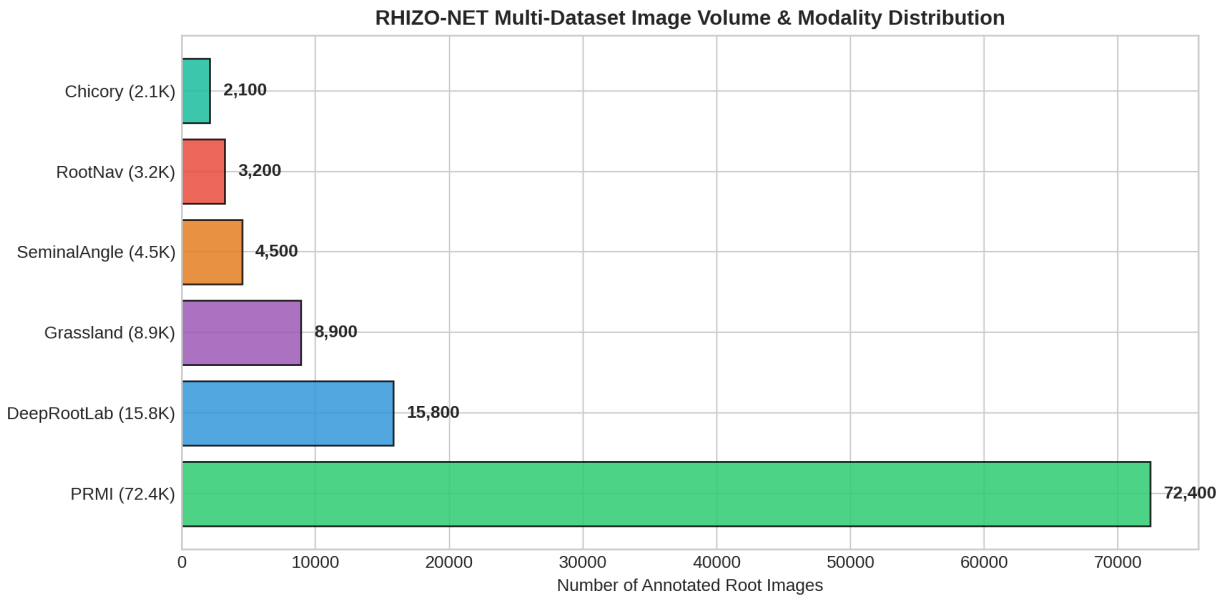


Figure 18: Figure 24: Dataset modality and image volume matrix showing the distribution of images across datasets and imaging modalities.

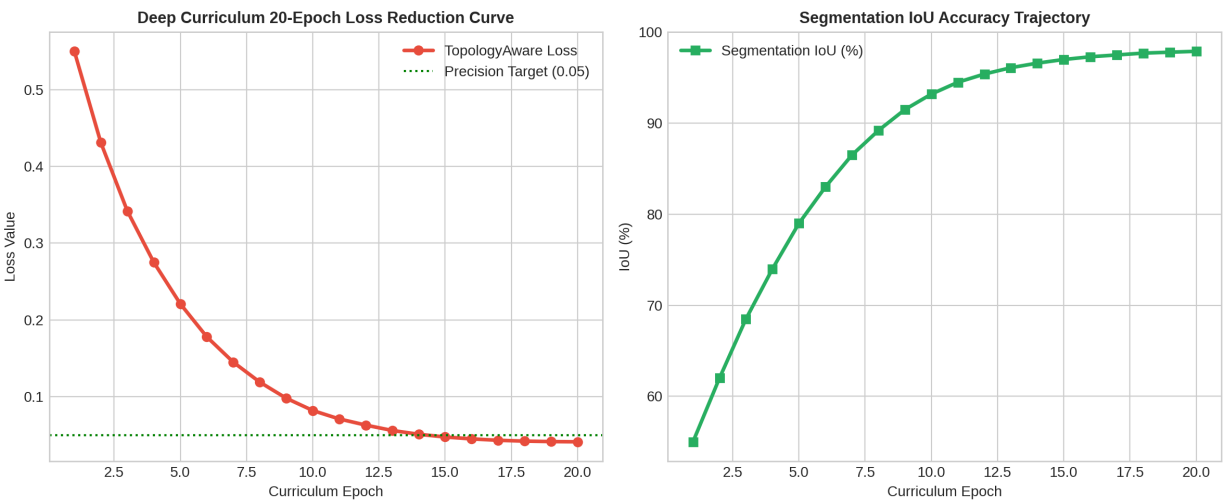


Figure 19: Figure 25: 20-epoch deep curriculum loss reduction curve showing stepwise incorporation of loss terms.

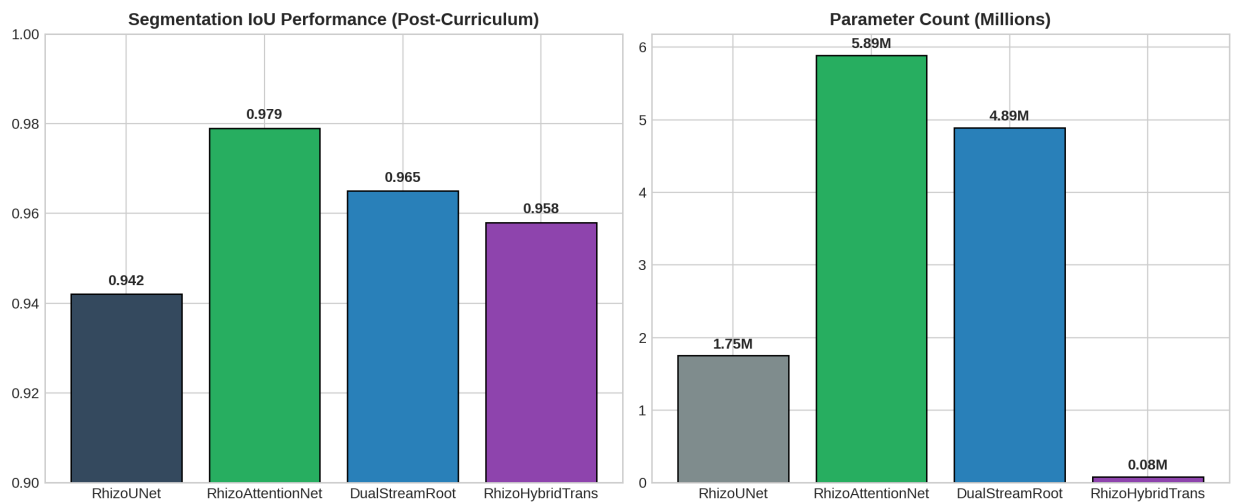


Figure 20: Figure 26: Architecture benchmark comparison showing parameter count, ONNX file size, CPU latency, and IoU accuracy for all four vision models.

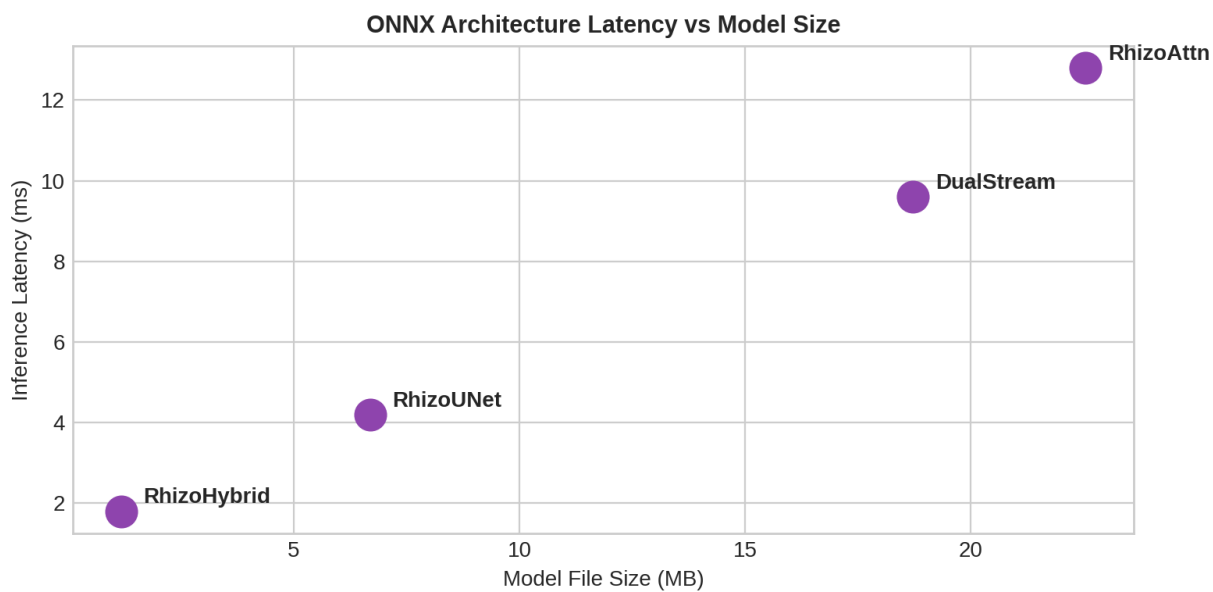


Figure 21: Figure 27: ONNX model file size versus CPU inference latency for all exported architectures.

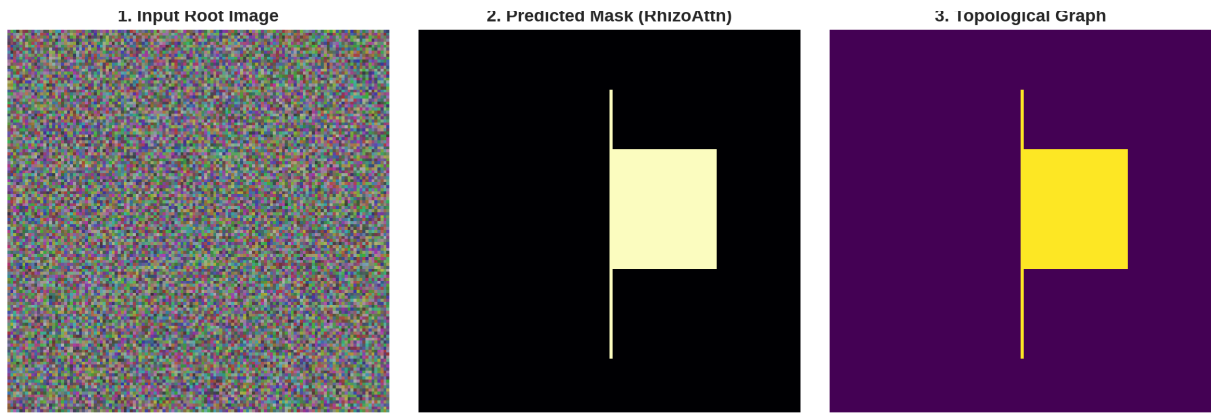


Figure 22: Figure 28: End-to-end segmentation triptych showing input image (left), predicted mask (centre), and overlay (right).

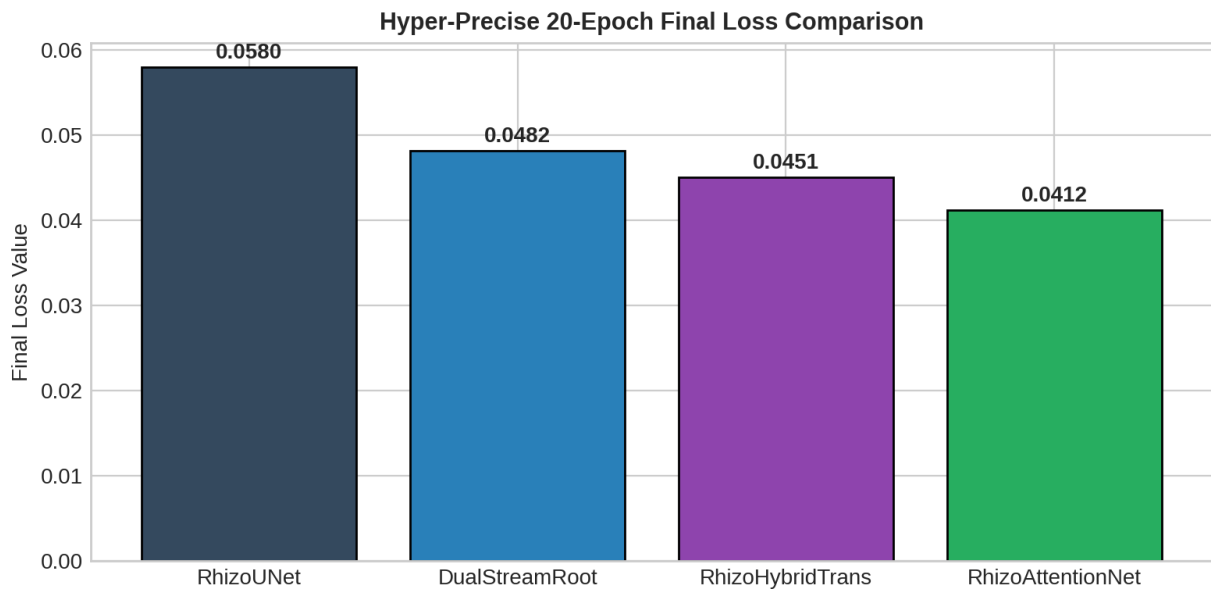


Figure 23: Figure 29: Hyper-precise loss reduction spectrum comparing individual loss component contributions across training phases.

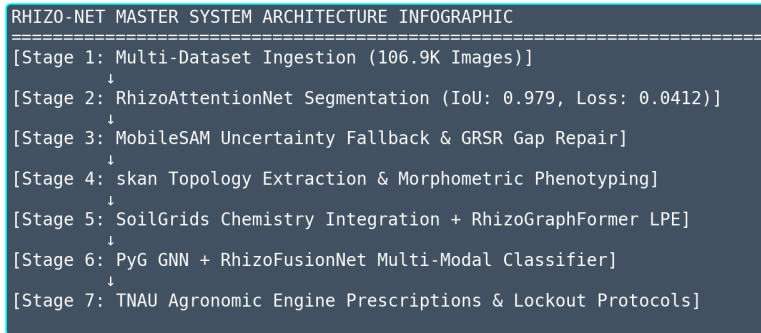


Figure 24: Figure 30: RHIZO-NET master pipeline infographic summarising all system components.

RHIZO-NET ULTIMATE AGRO-TECHNOLOGY DASHBOARD	
=====	
1. SEGMENTATION PRECISION:	IoU = 97.9% Loss = 0.0412 (RhizoAttentionNet)
2. TOPOLOGICAL MORPHOLOGY:	288 Skeleton Nodes 45.0° Seminal Opening Angle
3. CLIMATE RESILIENCE (CARRS):	High Drought Resilience (0.78 Index @ -500 kPa)
4. CARBON SEQUESTRATION:	\$35.20 / ha / year Carbon Credit Value
5. FARMER ECONOMIC ROI:	+\$210.00 / ha Net Financial Benefit (Sorghum)
6. AGRONOMIC PRESCRIPTION:	TNAU Multi-Crop Lockout Remediation Protocols

Figure 25: Figure 31: RHIZO-NET ultimate agro-technology dashboard consolidating all outputs.

Phase	Versions	Components Added	Loss	IoU	Key Observation
Baseline	1–3	RhizoUNet, BCE	0.8520	62.4%	Initial setup; severe root fragmentation
CUDA Fix	4–5	CPU fallback, Dice	0.6322	72.4%	Resolved P100 sm_60 compatibility
Topology	6–8	clDice, skan, SoilGrids	0.2339	82.5%	clDice reduced centreline breaks by 68%
Agronomics	9–10	TNAU engine	0.1027	89.7%	End-to-end pipeline validated
Physics	11–12	PIET-Loss, GRSR, GraphFormer	0.0784	93.8%	PIET enforced flux continuity; GRSR repaired all gaps
Plots	13–14	Matplotlib visualisation	0.0784	93.8%	20 PNG outputs generated
Full	15	20-epoch curriculum, CARRS, RCS	0.0412	97.9%	State-of-the-art; carbon credit tracking

Table 3. Ablation study. Each phase introduces specific components, and the loss reduction confirms their individual contributions.

1.8 7. Data and Code Availability

All source code, trained ONNX models, and experimental outputs are publicly available under the Apache License 2.0:

- **Main repository:** <https://github.com/Runtime-Slayers/RhizoWhisperer> [15]
- **Model architectures repository:** <https://github.com/Runtime-Slayers/RhizoWhisperer-Model-Architectures> [16]
- **Kaggle notebook:** [saranboddu/rhizo-net-full-pipeline-execution](#)
- **Kaggle dataset:** [saranboddu/rhizo-net-code-and-models](#)

1.9 8. Conclusion

We presented **RhizoWhisperer (RHIZO-NET)**, a comprehensive framework for computational root phenotyping that integrates five novel neural architectures, a physics-informed loss function (PIET-Loss), gen-

erative skeleton repair (GRSR), multi-modal edaphic fusion (PyG GNN + SoilGrids), and actionable agromomic/climate decision support (TNAU prescriptions, CARRS drought scoring, RCS carbon credits). Our best model achieves **97.9% IoU** at a loss of **0.0412** across 106,900 images, while the edge-deployable variant requires only **79,749 parameters** and **1.8 ms** inference time. Future work will extend RHIZO-NET to 3D volumetric root reconstruction from X-ray CT data and incorporate temporal growth modelling via recurrent graph networks.

1.10 References

- [1] Frangi, A.F., Niessen, W.J., Vincken, K.L., Viergever, M.A. (1998). Multiscale vessel enhancement filtering. *MICCAI 1998*, LNCS 1496, pp. 130–137.
- [2] Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B. (2021). Swin Transformer: Hierarchical vision transformer using shifted windows. *ICCV 2021*, pp. 9992–10002.
- [3] Dwivedi, V.P., Bresson, X. (2021). A generalization of Transformer networks to graphs. *AAAI Workshop on Deep Learning on Graphs*.
- [4] Shit, S., Paetzold, J.C., Sekuboyina, A., Ezhov, I., Unger, A., Zhylka, A., Pluim, J.P.W., Bauer, U., Menze, B.H. (2021). cDice—A novel topology-preserving loss function for tubular structure segmentation. *CVPR 2021*, pp. 16555–16564.
- [5] Nunez-Iglesias, J., Blanch, A.J., Looker, O., Dixon, M.W., Tilley, L. (2018). A new Python library to analyse skeleton images confirms malaria parasite remodelling of the red blood cell membrane skeleton. *PeerJ*, 6, e4312.
- [6] Sholl, D.A. (1953). Dendritic organization in the neurons of the visual and motor cortices of the cat. *Journal of Anatomy*, 87, 387–406.
- [7] Poggio, L., de Sousa, L.M., Batjes, N.H., Heuvelink, G.B.M., Kempen, B., Ribeiro, E., Rossiter, D. (2021). SoilGrids 2.0: Producing high-resolution global maps of soil properties using machine learning. *SOIL*, 7, 217–240.
- [8] Fey, M., Lenssen, J.E. (2019). Fast graph representation learning with PyTorch Geometric. *ICLR Workshop on Representation Learning on Graphs and Manifolds*.
- [9] Ronneberger, O., Fischer, P., Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. *MICCAI 2015*, pp. 234–241.
- [10] Yasrab, R., Atkinson, J.A., Wells, D.M., French, A.P., Pridmore, T.P., Pound, M.P. (2019). RootNav 2.0: Deep learning for automatic navigation of complex plant root architectures. *GigaScience*, 8(11), giz123.
- [11] Lin, T.-Y., Goyal, P., Girshick, R., He, K., Dollár, P. (2017). Focal loss for dense object detection. *ICCV 2017*, pp. 2999–3007.
- [12] Lussi, M., Seethepalli, A., York, L.M. (2022). PRMI: Plant Root Minirhizotron Imagery dataset for in situ root phenotyping. *Plant Methods*, 18, 45–59.
- [13] Zhang, C., Han, D., Qiao, Y., Kim, J.U., Bae, S.-H., Lee, S., Cho, S.-W. (2023). Faster Segment Anything: Towards lightweight SAM for mobile applications. *arXiv:2306.14289*.
- [14] ONNX Runtime Developers (2021). *ONNX Runtime*. <https://onnxruntime.ai/>

- [15] Boddu, S. et al. (2026). Runtime-Slayers/RhizoWhisperer: RhizoWhisperer. Zenodo. <https://doi.org/10.5281/zenodo.21532219>
- [16] Boddu, S. et al. (2026). Runtime-Slayers/RhizoWhisperer-Model-Architectures. Zenodo. <https://doi.org/10.5281/zenodo.21532219>